



Research article

Computation of tensile and compressive strengths of additively manufactured ABS material for automotive applications using ANN algorithms

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ABSTRACT

Experimental approaches and finite element analysis (FEA) are used to optimise the material extrusion (MEX) parameters for achieving effective tensile and compressive strengths of acrylonitrile-butadiene-styrene (ABS), which can lead to material waste and be costly. Additionally, techniques such as basic empirical models have failed to capture the complex non-linear interactions, resulting in inaccurate estimations. To address these challenges and compute the tensile and compressive strengths of additively manufactured ABS, this study develops novel artificial neural network (ANN) models using the Adam-optimised and Bayesian-regularised algorithms trained on an experimental dataset. Compared to traditional benchmarks, this study exhibited exceptionally low mean square error (MSE), root mean square error (RMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE) across hold-out and five-fold cross-validations. Industrial validations using the Taguchi L_{15} parametric combinations, i.e., print speed, layer height, infill density, bed temperature and nozzle temperature, yielded a coefficient of determination (R^2) above 0.90 for both properties. The case studies on brake-pedal and door-handles exhibited that the Adam-optimised and Bayesian-regularised ANN models predicted properties with MAPE below 1 %, reducing material wastage, development and lead times by 70 %, 60 % and 80 %, respectively. This study indicated the infill density as a critical factor influencing the strengths of the ABS material and thus, contributing towards achieving sustainable development goal (SDG) 9 by ensuring these cost-effective frameworks achieve effective strengths through optimised MEX-print parameters in additive manufacturing (AM) and automotive applications.

1. Introduction

Compared to conventional approaches, additive manufacturing (AM) slashes material waste, eliminates the need for tooling and moulding and contributes to greener manufacturing by enabling an economical and customised production without exorbitant costs [1,2]. This has revolutionised the manufacturing domain, boosting the global market cap to USD 20.37 billion in 2023 and is expected to hit a growth rate of 23 % by 2030. This demand is aided by an extensive use of AM in aerospace, defence, consumer goods, healthcare and automotive sectors [3–5].

Despite the AM processes offering a wide range of advantages, the part performance hinges tightly and is controlled by process parameters such as layer height, print speed, infill density, bed and nozzle

temperature [6,7]. Variations in these parameters enhance the mechanical properties by up to 60 % and sometimes even reverse their influence, especially on tensile and compressive strengths [6]. Although experimental studies have demonstrated optimal AM settings for ASTM standard polymers, their chronophagous and arduous nature has deterred adoption in industries [8,9]. Additionally, frequent cracks, distortions and delamination are observed in several tuned parameters, highlighting the need for more effective approaches [10,11]. Another severe limitation hindering the industry uptake is the failure of current standards to account for the printing parameters and anisotropic behaviour [10]. In the case of large-scale AM of polymers, thermal gradients and residual stresses are critical challenges observed [11].

To address these issues, material geometry is analysed and considered in another experimental approach, where comparative studies

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between virgin, recycled and fibre or adhesive embedded polymers are explored at different parameter settings [12]. Studies have shown that layer-height adjustment improved the tensile strength in one scenario, while the impact property decreased in another. This depicted the nonlinear and interacting effects of AM parameters [13–15]. Additionally, from the observations made, toolpath, void content and bed placement affect the material strength to some extent [16].

These failures and constraints in traditional experimental methodologies have paved the way to computational techniques such as finite element methods (FEM) and Response Surface methods (RSM) in optimising printing parameters [17–19]. Recent studies on fused filament fabrication (FFF) have used finite element approaches to simulate warpage in materials such as polyamide 12 and identified bed temperature as a critical factor [20]. Another study employing response surface methods (RSM) to optimise printing parameters in prosthetic sockets has demonstrated excellent correlation between experimentation and predictions [21]. However, these simulation-based studies require material models, meshes and larger computational times, which seems impractical. Additionally, its reliance on assumptions and boundary conditions not only leaves these techniques vulnerable but also critical in the industrial landscape [22–24].

Design of experiments (DoE) and RSM, in addition to Taguchi methods, can understand material interactions based on parameter combinations using factorial, fractional or orthogonal array designs [25]. Although these techniques are efficient, they require experimentation to validate every time as the number of factors increases, making them infeasible in larger application areas [25–29]. Some studies using RSM methods have demonstrated low-order polynomial models, enabling optimisation. However, approximations and non-linear parametric relationships could not capture complex data, resulting in failures [25,30].

Addressing these drawbacks, algorithm-based computational strategies are explored, where the use of genetic algorithms and swarm optimisation techniques for parametric studies resulted in deciding balanced and optimal print settings [31,32]. The key findings from the study reveal that the surface quality and mechanical properties have considerably improved [31,32]. Another approach using the grey wolf optimisation algorithm has successfully predicted the optimal print settings, effectively balancing the multi-objective problems, cost and quality constraints [33]. Although these algorithm-based approaches offer fine-tuned solutions, the complexity involved makes it chronophagous and arduous to implement [32,34].

Demanding applications in the automotive sector, approaches to solving various challenges are increasing. Some studies that used artificial neural networks (ANNs) predicted and optimised internal combustion engine characteristics, reducing experimentation time and costs [35]. Tests with these standards are performed to assess the strength of aluminium composites that exist in automobiles [36,37]. Similarly, in studies concerning the propulsion system calibration, the ANNs can speed up experimental processes by forecasting control parameters, significantly reducing calibration time [38]. These diverse applications highlight the versatility and potential of ANNs in improving various aspects of automotive engineering.

However, due to limited studies related to the use of ANNs in optimising the strengths of polymer composites based on 3D-printing parameter combinations, especially for ABS, in various automotive applications and the limitations and challenges in various traditional and computational approaches, there is a dire need for modern data-driven techniques in understanding the complex non-linear relationships more effectively and predicting properties without any extensive experimental overheads [39,40]. This study aims to address the current industrial demand of AM in the automotive sector with the following novel objectives;

1. To develop, train, validate and test the Adam-optimised and Bayesian-regularised ANN models.

2. To obtain performance metrics of both the ANN models and draw a comparative analysis with the decision tree (DT), simple vector machines (SVM), random forest (RF), XGBoost (XGB), support vector regression (SVR), K-nearest neighbours (KNN) and Ada Boost (ADAB) approaches.
3. To assess the prediction accuracy through the coefficient of determination (R^2) and validate industrial implications through the Taguchi L_{15} method and its applicability through case studies.
4. To evaluate the effect of printing parameter variations on ABS material's tensile and compressive strength through the Pearson correlation analysis (r) and draw the cost-benefit and sustainability assessment of the models for effective industrial use.

By harnessing the strengths of Adam-optimised and Bayesian-regularised algorithms, this novel approach aims to offer rapid, sustainable and industrially ready models in predicting tensile and compressive strengths of ABS material based on printing parameters employed. This approach not only offers high-precision manufacturing but also lays a versatile foundation for scaling 3D printing towards sustainable automotive production ecosystems.

2. Methodology

The methodology of this study encompasses the material, specimen standards, the printing process, parameters and experimental design employed. Additionally, the systematic procedure in preparing the training dataset and its validation through bibliometric analysis is elucidated. The development of the Adam-optimised and Bayesian-regularised ANN models and the rationale for selecting these strategies are explained. The need and effectiveness of performance metrics and validation in evaluating the prediction accuracy are examined.

2.1. Material used

To ensure an optimum performance of the ABS, the structural properties of importance to engineering uses are considered. The typical specific gravity is 1.05 g/cm^3 with an approximate equivalent range of 1.04 g/cm^3 to 1.07 g/cm^3 [41]. Young's modulus is in the range of 1500–2500 MPa, which is a reflection of moderate stiffness [42]. The relative impact of notched Izod impact strength is typically measured at 11.6 kJ/m^2 , which serves as a good indicator of toughness [41]. The determining heat-deflection temperature (HDT) of 80.5°C , along with a linear coefficient of thermal expansion of approximately $90 \times 10^{-6} / ^\circ\text{C}$, is considered [41,43]. These properties, as tabulated in Table 1, form the basis of the choice of ABS as a long-lasting, dimensionally stable polymer suitable for automotive engineering applications. These ABS filaments are procured from a vendor (Vyapti Tecnica Solutions LLP, Bengaluru, India) and stored in controlled vacuum-sealed containers at a room temperature of 18°C to 26°C , ensuring the hygroscopic nature and moisture absorption do not affect the printing quality and adhesion [44].

The spools of filaments are placed in a desiccator filled with air to a relative humidity of 20 % at $23^\circ\text{C} \pm 1^\circ\text{C}$, where they are dried in a vacuum oven at 60°C for 24 h. The drying process performed follows the ASTM D6869–17 to minimise potential moisture variability during extrusion [45].

Table 1
Key properties of ABS material.

Property	Value	Reference
Density (specific gravity)	1.05 g/cm^3	[41]
Notched impact-strength	11.6 kJ/m^2	[41]
Heat deflection temperature	80.5°C	[41]
Young's modulus	1500–2500 MPa	[42]
Coefficient of thermal expansion	$90 \times 10^{-6} \text{ m}/^\circ\text{K}$	[43]

2.2. Specimen standards and 3D printing process

To ensure standardisation in testing and experimentation, studies relevant to CAD models complying with ASTM D638 and ASTM D695 standards are considered. These models are printed in an XY plane at an ambient room temperature of $25^{\circ}\text{C} \pm 1^{\circ}\text{C}$, maintaining consistency in printing. The entire printing process is carried out through a material extrusion (MEX) 3D printer named PRATI_AP (OEM Prati contract manufacturing - Vyapti Tecnica Solutions LLP, Bengaluru, India) with a build volume of $500\text{ mm} \times 500\text{ mm} \times 500\text{ mm}$ and 2 print heads (one swappable to pallet printing), under controlled environmental conditions (room temperature of 22°C to 26°C) that meet industry standards, ensuring precise control over printing parameters. The calibration of all temperature sensors in the MEX 3D printer (thermocouples with a $\pm 0.1^{\circ}\text{C}$ accuracy) reduces systematic errors in measurements of bed temperature and nozzle temperature through drift checks. Any deviations exceeding $\pm 0.2^{\circ}\text{C}$ trigger recalibration. The maximum temperature for heating the filaments for extruders is 300°C , with a filament diameter of 1.75 mm being required.

2.3. Printing parameters studied

To ensure smooth training, testing and industry validations, a total of 5 key printing parameters are carefully studied and considered in this study, with respective ranges detailed in Table 2. Each of these parameters is evaluated and its impact on properties is analysed. ABS normally requires higher extrusion temperatures to achieve strong layer adhesion; thus, a combination of nozzle (hotend) temperatures of 200°C to 250°C is used, including the commonly quoted ABS range (approximately $200\text{--}250^{\circ}\text{C}$) [46–49]. The range of the heated bed temperatures is between 50°C and 110°C , with values in both moderate and high domains that are advised in order to enhance ABS bonding and avoid warping [46–49]. The infill density is deliberately modified between 20 % (lightweight) and a complete 100 % (solid), thus representing the entire range of part solidity [50–53]. The layer height is adjusted to 0.2 mm, 0.5 mm and 0.8 mm, to explore the fine-scale to coarse-scale resolutions; this is arrived at because literature showed that there is a significant variation in strength at this scale of resolutions. Since 0.1 mm layer height is abundantly explored in studies and the incremental benefit in mechanical properties is minimal and the difference in print time is significantly longer compared to other layer heights, the consideration of 0.1 mm in this study is thus excluded [54–57]. These parameters are followed to print the material at a speed of $10\text{--}70\text{ mm/s}$ [58–60]. These values of the parameters (Table 2) therefore encompass both standard and extreme MEX conditions.

2.4. Experimental design technique

Various Taguchi techniques, such as Taguchi L_9 , L_{16} and L_{27} , efficiently investigate multiple factors at different levels [61–64]. However, to ensure precise industry-relevant parameter combinations, maintain the highest testing standards and quality through minimising time consumption, cost and resources, this study explores the Taguchi L_{15} quasi-orthogonal array [65]. It maintains the statistical power and efficiency by capturing the effects with fewer trials. Five factors at three levels are fitted, but the experiments are made smaller so that only 15

experiments are run and main effects are captured [65,66]. The three levels of layer heights (0.2 mm, 0.5 mm and 0.8 mm) included, in particular, a wide range of deposition thicknesses; the previous results indicate that the strength is maximised under small layer heights (approximately 0.2 mm) and diminishes at large heights (approximately 0.5 mm and 0.8 mm) [54–57]. Layers at 0.1 mm (which have found utility with very fine detail modelling) are omitted since their use results in print times that are exponentially longer with only small increase in strength and are less relevant in general auto part prototyping [67,68]. Compared to traditional approaches such as one-factor-at-a-time (OFAT), full factorial or fractional factorial design, Plackett-Burman or central-composite design (CCD), Box-Behnken design, latin square or simple random design, this technique ensures a balance between two-factor interactions against resource constraints [69–72]. Overall, L_{15} with the mentioned factor levels offers a fair coverage of the parameter space, which is consistent with what the literature on the methodology of Taguchi design for optimisation of manufacturing systems exhibits. These parameter combinations, as listed in Table 3, are later used for predicting tensile and compressive properties using the Adam-optimised and Bayesian-regularised ANN models. For each of these 15 combinations, the experimental tensile and compressive strengths are used for comparative analysis.

2.5. Training dataset

The training dataset comprises 383 tensile and compressive strengths, compiled through an extensive literature survey of peer-reviewed studies, industry articles and tested reports [73].

2.5.1. Selection criteria and quality control

To maintain consistency, studies focusing on virgin, commercial-grade or industrially refined ABS material/filaments are considered. The tensile or compressive strengths on ASTM/ISO standard specimens

Table 3
Taguchi L_{15} parameter combinations.

Combinations	Nozzle temperature ($^{\circ}\text{C}$)	Bed temperature ($^{\circ}\text{C}$)	Infill density (%)	Print speed (mm/s)	Layer height (mm)
Combination 1	200	50	20	10	0.2
Combination 2	200	80	60	40	0.5
Combination 3	200	110	100	70	0.8
Combination 4	225	50	20	40	0.5
Combination 5	225	80	60	70	0.8
Combination 6	225	110	100	10	0.2
Combination 7	250	50	60	10	0.2
Combination 8	250	80	100	40	0.8
Combination 9	250	110	20	70	0.5
Combination 10	200	50	100	40	0.5
Combination 11	200	80	20	70	0.8
Combination 12	200	110	60	10	0.2
Combination 13	225	50	100	40	0.2
Combination 14	225	80	20	70	0.5
Combination 15	225	110	60	10	0.8

Table 2
Key printing parameters and their range.

Parameters	Range	References
Nozzle temperature	$200\text{--}250^{\circ}\text{C}$	[44–47]
Bed temperature	$50\text{--}110^{\circ}\text{C}$	[44–47]
Infill density	20–100 %	[48–51]
Layer height	0.2–0.8 mm	[52–55]
Print speed	10–70 mm/s	[56–58]

under calibrated testing procedures and parameters are prioritised.

The entire raw data is further subjected to quality control, where outliers, deviations and inconsistencies are excluded. Finally, a balanced dataset of 383 parametric combinations and properties is compiled [73].

2.5.2. Dataset splitting and normalisation

The compiled dataset is split into training (70 %), testing (15 %) and validation (15 %) subsets, which are maintained the same for hold-out and k-fold cross-validation techniques in both models. A min-max scaling process is carried out to eliminate any scale differences between printing parameters, ensuring stability between input and output variables. This leads to the robustness and generalisability of ANN models.

Despite the curated 383 sample dataset having been checked against outliers via omitting observations that are greater than the removal point of 3 sigma and having found that all inputs exhibit insignificant skewness ($|\text{skew}| < 0.5$), there still may be limitations regarding the heterogeneity of the sample, given that it is found in the literature studies of the dataset [73]. The studies indicate that the infill density occurred more frequently in 20 % and 100 % and so a histogram of the expected distribution has an over-representation at these, which is likely to impact the performance of the models in the middle-range prediction.

2.5.3. Co-occurrence analysis by study keywords

To select the most effective parameter combinations, a keyword co-occurrence analysis is conducted on 228 peer-reviewed studies obtained from reputable indices such as Scopus and Web of Science. This validates the factor space, the selection of parameters and the parameter ranges.

To extract relevant studies, keywords such as additive manufacturing, fused deposition modelling, acrylonitrile-butadiene-styrene (ABS), mechanical properties, tensile strength and compressive strength are used. Further, a cleaning protocol is applied, which includes merging, standardisation, hyphenation, and stop-word removal. The resulting dataset of keywords is saved in .ris format and fed to the VOSviewer software, extracting the co-occurrence analysis by study keywords.

The importance of neural networks, process optimisation and mechanical properties highlights the suitability of neural network models in understanding the non-linear behaviours. Similarly, clustering the experimental design, the Taguchi method and parameter interactions validates the approach (as shown in Figure A in Annexure A).

However, the sparse co-occurrence between the Bayesian-regularised, Adam-optimised and ABS material has revealed the potential research gap, motivating the study choice (as shown in Figure A and Figure B in Annexure A). These insights from co-occurrence analysis play a crucial role in the parameter selection and thus in preparing the training dataset.

2.6. Development of ANN models

Based on the studies and insights from the co-occurrence analysis, the Adam-optimised and Bayesian-regularised ANN models are embedded with certain common foundational choices. The network architecture in both models has five neurons in the input layer: the nozzle temperature, bed temperature, infill density, layer height and printing speed. Similarly, the output layer comprises two neurons: the tensile and compressive strengths. The output layer is designed to predict the properties based on the input parametric combinations.

2.6.1. Adam-optimised artificial neural network model

To ensure proper capturing of non-linearity, the Adam-optimised ANN model employs a feed-forward network, a z-score normalisation for five inputs and a hyperbolic tangent activation function with two connected layers. The two output layers used for strength predictions are connected with a regression layer for computing the mean squared error (MSE) losses.

The implication of 1000 epochs and a batch size of 32 samples ensures smooth convergence in training and allows efficient learning. Additionally, from initial sweeps across 10^{-4} to 10^{-2} , a learning rate of 1×10^{-3} is fixed for rapid learning and avoiding overshooting.

With these configurations, a training is executed and the real-time validation losses are depicted on the charts. This ensures immediate action on failures to converge, such as debugging or adjusting hyper-parameters. The Fig. 1 represents the schema of the Adam-optimised ANN model developed in this study.

2.6.2. Bayesian-regularised artificial neural network model

A feedforward network topology with two hidden layers, each with 20 neurons, is employed. The hyperbolic tangent sigmoid and the linear activation functions are used in the hidden and output layers, respectively. Fig. 2 represents the schema of the Bayesian-regularised ANN model developed in this study.

To ensure proper training, a maximum of 1000 epochs and a batch size of 42 samples are set. However, if no improvements are made for six consecutive epochs, then an early stopping protocol is made active, halting the training. To ensure smooth training of the model, this protocol activation is automated after six continuous hits.

The Bayesian algorithm is optimised such that the network weights and bias regularisation parameters are adjusted automatically, balancing data fitness and network complexity. Further, the network is trained with an enabled GUI monitoring, ensuring interactive observation. With these configurations and methodology, the training is executed.

2.7. Performance metrics

To evaluate the training and prediction accuracy of the Adam-optimised and Bayesian-regularised ANN models, quantitative performance metrics are used under hold-out and k-fold cross-validation strategies.

To validate the accurate training, MSE is used as a loss function, as shown in Eq. 1, driving the network to reduce the outliers. Squaring the individual errors obtained, MSE penalises large deviations, which is crucial for safety-critical applications to avoid mispredictions [74]. To understand how many units the predicted properties have deviated during testing, the root mean square error (RMSE), as shown in Eq. 2 is used. It brings the squared error penalty to the same target units, ensuring interpretability and helping set tolerances in parameter settings [75].

As ABS behaviour is a factor of replication of strengths, it is crucial to know whether the standard property range is achievable or not. This is achieved via mean absolute error (MAE) as shown in Eq. 3, helping maintain the quality, robustness and compliance with specifications. However, to find out tensile and compressive strengths based on parametric variations, switching between materials and different strength levels is an intuitive process and thus, relative accuracy is crucial in production settings [76]. Mean absolute percentage error (MAPE), as shown in Eq. 4, expresses errors in percentage terms, facilitating benchmarking against other materials [77].

To indicate the level of prediction accuracy when compared with experimental values based on L_{15} parametric combinations, the coefficient of determination (R^2), as shown in Eq. 5, reliably tracks the trends and variations in predictions [78]. A strong R^2 ensures the excellent prediction accuracy of the ANN model, while a low coefficient signifies errors and real-world failure.

The following formulations are employed, defining the MSE, RMSE, MAE, MAPE and R^2 functions.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

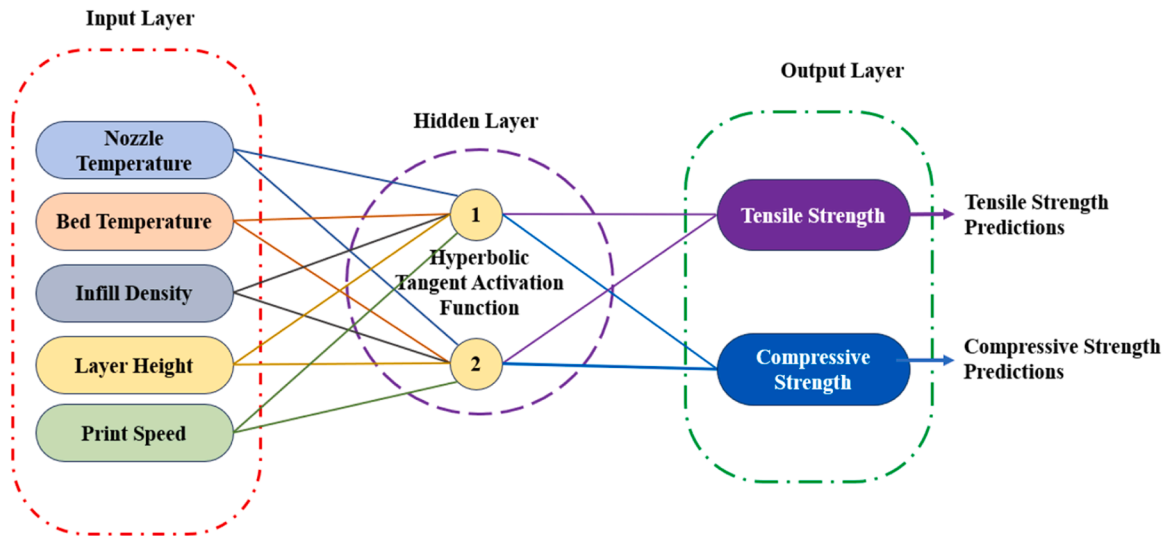


Fig. 1. Schema of the Adam-optimised ANN model.

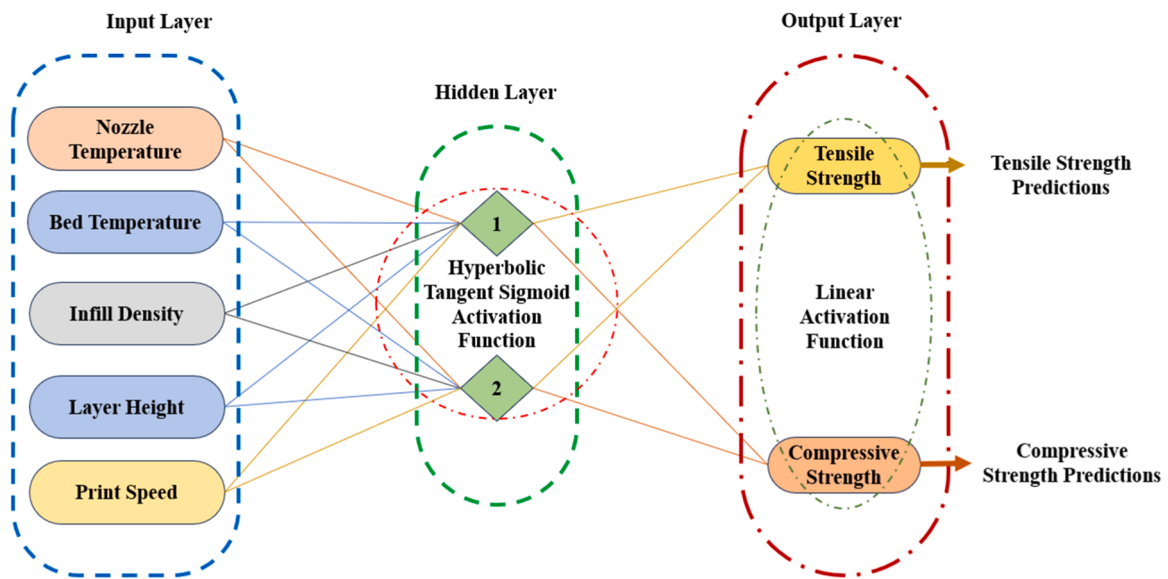


Fig. 2. Schema of the Bayesian-regularised ANN model.

$$RMSE = \sqrt{MSE} = \sqrt{\left(\frac{1}{n} \sum_{i=1}^n (y_i - y_a)^2\right)} \quad (2)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y_a| \quad (3)$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - y_a}{y_i} \right| \quad (4)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - y_a)^2}{\sum_{i=1}^n (y_i - y_b)^2} \quad (5)$$

Where; y_i = true values; y_a = predictions; y_b = mean of true values; n = number of samples

With a precise combination of these five performance metrics, the study captures both typical and delicate errors, assessing the developed ANN models directly for industry relevance, accuracy and standards.

2.7.1. Hold-out cross-validation

To carefully assess the generalisation of both models, stratification of the training dataset is done and it is ensured that the property distribution closely mirrors the overall dataset. This helps in preventing skewed evaluation results. Further, model-compatible normalisation strategies are used, eliminating the further reprocessing of predicted values or manual reverse-scaling needs, adding to the novelty in both models.

2.7.2. k -fold cross-validation

A stratified 5-fold split is used in the Adam-optimised and Bayesian-regularised ANN models to ensure uniform sample distribution across folds. Using the fold-specific training subsets, the Adam-optimised algorithm leverages automatic normalisation of the input and output values, while the Bayesian-regularised model is re-normalised, if needed. While the Adam-optimised approach uses identical settings, the Bayesian-regularised ANN model uses retraining with early stopping. Finally, the predicted values are reverse-scaled in each fold in both models, ensuring no post-processing or scale adjustment of data is required.

2.8. Predictions and validation

To ensure industrial applicability of the models, Taguchi L_{15} quasi-orthogonal array parametric combinations are employed to evaluate tensile and compressive strengths. The results obtained are used for comparative analysis and thus, the prediction accuracy of both models is evaluated using the coefficient of determination (R^2).

3. Results and discussion

3.1. Training and validation

During the initial training phases, both models demonstrated an excellent learning capacity for a complex, nonlinear relationship between printing parameters and properties, rapidly minimising their loss functions. The Adam-optimised algorithm successfully achieved an MSE of 0.0523 and 0.0586 for both tensile and compressive strengths, across hold-out and 5-fold cross validations. Similarly, the Bayesian-regularised approach achieves an exceptionally low and uniform MSE of 0.0001 under hold-out cross-validation for tensile and compressive strengths. Under 5-fold cross-validation, MSE values of 1.7543 and 1.6458 are obtained, respectively.

From Table 4, the RMSE and MAPE for tensile and compressive strengths in the Bayesian-regularised approach have outperformed those of the Adam-optimised model. In the case of MAE, the Bayesian-regularised model has shown considerable weakness compared to the Adam-optimised model during the 5-fold cross-validation process. This difference can be explained on the basis that the early-stopping criterion is reached in fold 4, when some of the most peculiar parameter sets are used, including 0.8 mm layer height at 70 mm/s and the model is stressed to a point where it is stopped earlier in training. Model complexity penalties prevail in this set of parameters via higher out-of-sample variance. From the learning curves observed in both the models, it is clear that the Bayesian-regularised model started triggering early stoppings at around epoch 600, while the Adam-optimised model secured convergence around epoch 400, beyond which balancing of weight decay and marginal gains is noted till epoch 1000.

From the performance metrics depicted in Table 4, both models have demonstrated typical magnitudes for ABS strengths, indicating a realistic fit. With exceptional performance and effective regularisation, both models offer an effective approach in predicting properties. However, with a few discrepancies, unlikely to arise due to expected noise, further industry-relevant validations are provided over a Taguchi L_{15} parametric combination.

3.1.1. Hold-out cross-validation

With a learning rate of 1×10^{-3} for a batch size of 32 at epoch 1000, the Adam-optimised ANN model successfully attained uniform training, while the Bayesian-regularised ANN model experienced a tight regime, as reflected in the performance metrics detailed in Table 4. With a single central bin and the vast majority of errors falling within a small window of -1.3×10^{-4} to 1.4×10^{-4} , the histogram corroborates exceptional accuracy (as shown in Figure C in Annexure A).

3.1.2. k-fold cross-validation

Stratified five-fold cross-validation is performed to ensure the stability of Adam-optimised and Bayesian-regularised ANN models. The larger swings and gradual decrease in tensile and compressive strengths as observed in Fig. 3 are due to subtle variations in specimen standards and geometry, reflecting the authenticity of the model's generalisation. Conversely, the Bayesian-regularised ANN model, as shown in Fig. 4 occasionally produces outliers with excellent average performance, depicting a nonlinear relation between parameter combination and mechanical behaviour.

Since RMSE amplifies the MSE behaviour, the Adam-optimised ANN model underscores the test subsets that use higher print speeds or higher layer heights, thus challenging the properties and yielding larger squared errors. On the contrary, although the RMSE is imperceptible and under 0.02 during the first fold, it is corrected in the next folds, exhibiting the model's stability in training and testing (as shown in Figure D and Figure E in Annexure A).

As observed in the third fold from k-fold cross-validation, the lowest MAE depicts the easiest prediction possibility for the model characterised by its parameter combinations. Alternatively, the fourth fold has demonstrated the highest errors, challenging the model's generalisation due to the extreme changes in parameters such as infill densities or bed temperature.

The sharp increase in the MAE in the fourth fold also challenges the early-stopping protocol applied, which decreased the effective number of training epochs on that particular fold. Concisely, early stopping led to the network halting before the local optimisation of the fourth fold had provided enough local minimisation of the network, resulting in increased out-of-sample absolute errors even though the MSE and RMSE are held at low levels on average. The sensitivity check through re-training of a particular fold shows that the anomaly is, in part, an artefact of the stopping criterion instead of a genuine failure of the Bayesian regularised model.

Although the average MAE for tensile and compressive strengths is 5.4×10^{-3} and 5.6×10^{-3} , the gradual decrease in MAE reduces the tensile and compressive strengths that occur during the first and second folds. The potential reason for these variations is the specimen geometry and corresponding parameter regimes. From the third fold, the MAE plummets in both the tensile and compressive strengths, signifying the model's regularisation and capability in handling complex datasets (as shown in Figure F and Figure G in Annexure A).

In production settings where worst-case absolute errors are relevant to safety or performance margins, this behaviour would suggest reporting per-fold MAE and epoch counts to allow practitioners to understand fold-level robustness. Additionally, using a Bayesian model in heterogeneous parameter spaces, a conservative early-stopping strategy, or repeated training runs per fold would ensure that folds with rare/edge parameter combinations receive sufficient optimisation. Adherence to these measures may minimise the risk of unexpected absolute errors in industrial predictions caused by fold-specific early stopping.

The steady decrease in MAPE for tensile strength ensures the incremental drop in errors, validating training accuracy and predicting the property accurately. In contrast, the slight increase in errors during the fourth fold in compressive strengths signifies the possibility of existing

Table 4

Performance metrics for the Adam-optimised and Bayesian-regularised ANN models with hold-out and k-fold cross-validation.

Optimisation algorithm	Cross-validation strategy	Mechanical property	MSE	RMSE	MAE	MAPE
Adam-optimised	Hold-out	Tensile	0.0523	0.2031	0.1601	0.850
		Compressive	0.0523	0.2516	0.1973	0.840
	5-fold	Tensile	0.0586	0.2126	0.1541	0.740
		Compressive	0.0586	0.2661	0.1931	0.720
Bayesian-regularised	Hold-out	Tensile	0.0001	0.0100	0.0020	0.010
		Compressive	0.0001	0.0077	0.0019	0.010
	5-fold	Tensile	1.7543	0.0125	5.5000	0.030
		Compressive	1.6458	0.0120	5.5263	0.025

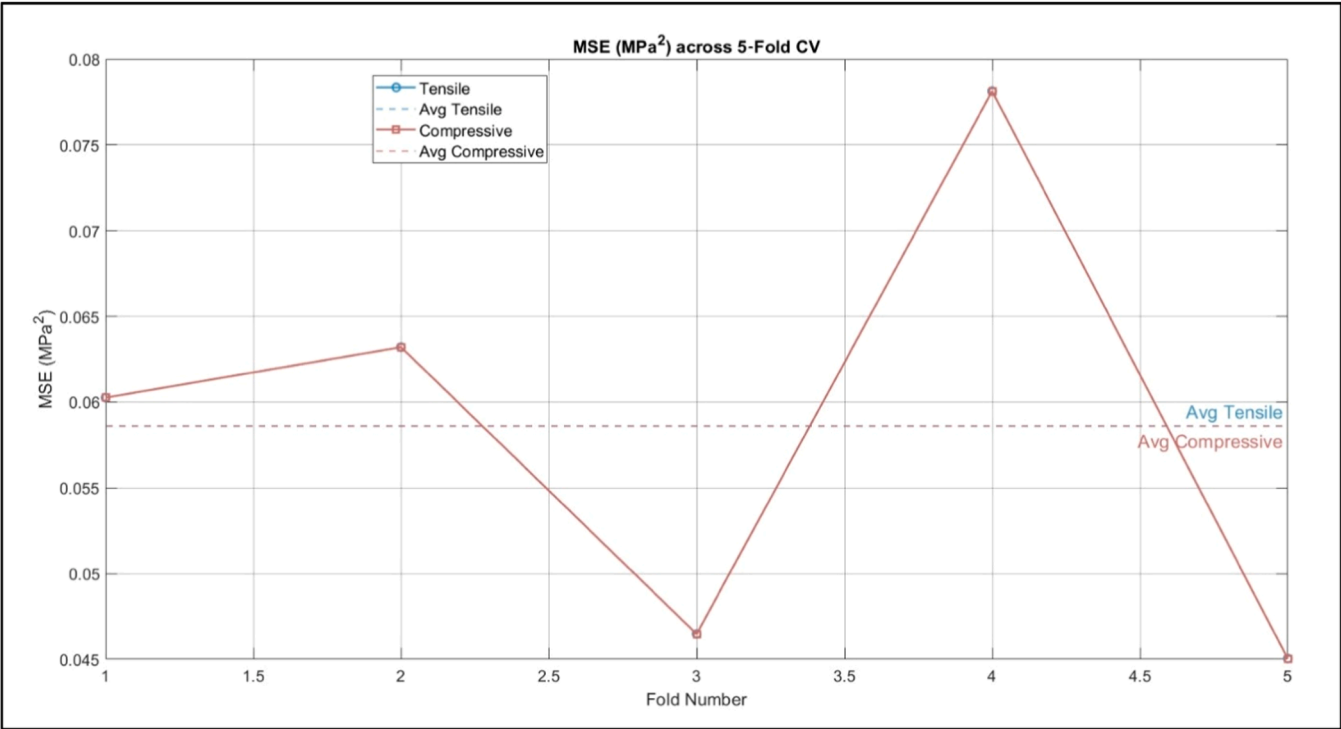


Fig. 3. Mean square error from k-fold cross-validation for the Adam-optimised ANN model.

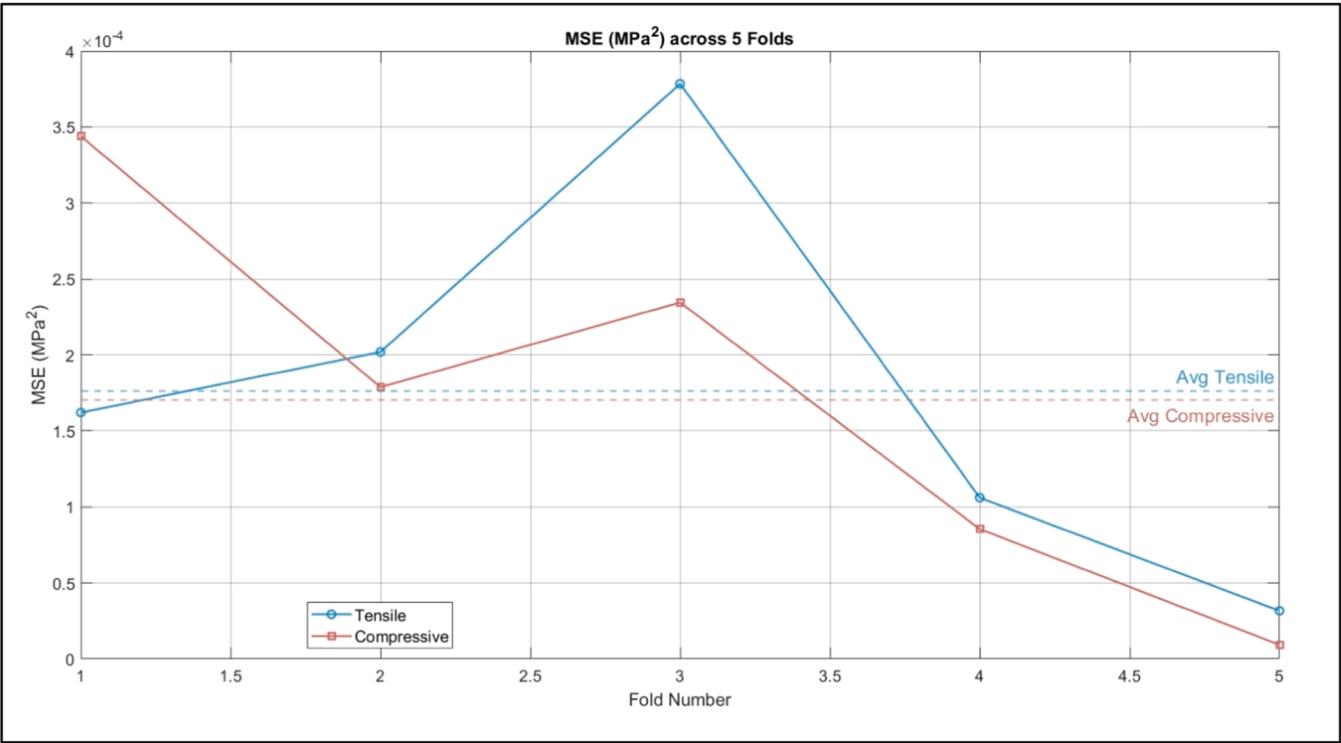


Fig. 4. Mean square error from k-fold cross-validation for the Bayesian-regularised ANN model.

variability in compressive testing. On the contrary, the rapid decline in MAPE from the second fold exhibits the Bayesian model’s exceptional capacity in handling complex parameters and material properties. As the model aligns with data characteristics, the near-zero error is obtained as reflected in the average MAPE for tensile (0.024) and compressive (0.032) strengths (as shown

in Figure H and Figure I in Annexure A).
3.1.3. Cross-fold synthesis and material implications
From the third fold’s performance in both models, it is evident that the training set employed aligns closely with the printing parameter combination where the ABS material’s interlayer bonding and

crystallinity are more stable. This is due to the use of mid-range nozzles and bed temperatures (ranging between 200°C to 220°C and 50°C to 70°C).

On the contrary, the challenging region, depicted in the form of a fourth fold, represents errors due to maximum layer heights or nozzle temperatures employed. These configurations lead to poor material deposition and adhesion, resulting in material failure, as reflected in the model predictions as well.

However, following successful training and validation, the lower RMSE and MAPE confirm the Adam-optimised ANN model's capability in capturing the complex relationship between parameters and ABS's tensile and compressive strengths. Additionally, the model's consistency illuminates the specific parameter combinations where material property complexity is heightened.

From the observations made in the higher folds 1 and 2 of MAE and MSE to the errors observed in folds 4 and 5, the Bayesian-regularised algorithm's capability to fine-tune the weights and constrain prediction results in regularisation convergence is validated. To identify the challenging regimes, the parameter combinations in the fold 3 subset of MSE and RMSE deviate from the model's expectations and their corresponding tensile and compressive strengths.

The near-experimental accuracy is obtained from the predictions of the Bayesian-regularised ANN model, ensuring its applicability to micro-precision in the prediction of tensile and compressive properties. While the folds increase, the errors plummet, implying that the model can be industrially applied, deploying its effectiveness with evolving process conditions.

3.1.4. Comparative analysis of performance metrics

To ensure the authenticity of optimum performance of the Adam-optimised and Bayesian-regularised ANN models, a comparative analysis is drawn with the various machine learning algorithms from the literature, as depicted in Table 5 and Table 6. The Adam-optimised ANN model is highly reliable for compressive strengths and is ideal for iterative design processes where parameter selection and optimisation are critical.

While the Bayesian-regularised ANN model proves to be a powerful tool for MEX process optimisations, its validity in reducing experimentation and waste generation is true. This makes the model valuable in industry settings.

The implication of the decision tree-based approach by Ahmet Burak Tatar et al. (2025) [79] to predict the tensile strength of a composite material exhibited an MSE, RMSE and MAE of 28.339, 5.3234 and 3.7534, respectively. These error rates are much higher than the ANN approaches in this study. Similarly, another study by Amadi Gabriel Udu et al. (2024) [80] using the same approach for compressive strength predictions on a similar composite material yielded a MSE and MAE of 0.0114 and 0.0324, which is, although exceptional, yet still lower compared to the Bayesian-regularised ANN model's hold-out cross-validation approach as shown in Table 6. The vast contrast in the

Table 5

Summary of comparative analysis of performance metrics for tensile strength predictions.

Model used	MSE	RMSE	MAE	MAPE
Adam-optimised (Hold-out)	0.0523	0.2031	0.1601	0.850
Adam-optimised (5-fold)	0.0586	0.2126	0.1541	0.740
Bayesian-regularised (Hold-out)	0.0001	0.0100	0.0020	0.012
Bayesian-regularised (5-fold)	1.7543	0.0125	5.5000	0.030
Decision tree (DT)	28.339	5.3234	3.7534	6.69
Simple vector machines (SVM)	16.938	4.1156	2.6383	-
Random forest (RF)	21.688	4.6517	3.0317	-
XGBoost (XGB)	1.2795	1.1312	3.1867	19.8537
Support vector regression (SVR)	1.7538	1.3243	4.5836	25.3681
k-Nearest Neighbours (k-NN)	0.0020	0.6644	0.0146	-
AdaBoost (ADAB)	0.0006	0.5598	0.0139	-

Table 6

Summary of comparative analysis of performance metrics for compressive strength predictions.

Model used	MSE	RMSE	MAE	MAPE
Adam-optimised (Hold-out)	0.0523	0.2516	0.1973	0.840
Adam-optimised (5-fold)	0.0586	0.2661	0.1931	0.720
Bayesian-regularised (Hold-out)	0.0001	0.0077	0.0019	0.010
Bayesian-regularised (5-fold)	1.6458	0.0120	5.5263	0.025
Decision tree (DT)	0.0114	0.0443	0.0324	-
Simple vector machines (SVM)	0.3980	15.46	8.70	10.81
Random forest (RF)	0.0113	0.0511	0.0812	0.04
XGBoost (XGB)	0.0071	0.9605	0.0252	-
Support vector regression (SVR)	0.0098	-	0.0860	-
k-Nearest Neighbours (k-NN)	0.0060	3.9694	0.0247	0.05
AdaBoost (ADAB)	0.0079	5.099	0.0036	-

decision tree model's performance for tensile and compressive strengths, as seen in the comparative analysis, signifies the approaches' limited applicability.

In the case of simple vector machines, a study by Ahmet Burak Tatar et al. (2025) [79] exhibited the MAE of 2.6383 for tensile strength, in contrast to the 8.70 for compressive strength obtained by Prashant Kumar et al. (2023) [81]. From Table 5, it is clear that the SVM approach is well-suited for tensile strength predictions, as justified by the error rate of 15.46 obtained in the compressive strength study compared to 4.1156 obtained by Prashant Kumar et al. (2023) [81] for the tensile strength study. In comparison to the metrics of the SVM approach for tensile and compressive strengths, both Adam-optimised and Bayesian-regularised ANN models tend to perform exceptionally well, as depicted in Table 5 and Table 6.

Ahmet Burak Tatar et al. (2025) [79] in their approach on the random forest algorithm to predict tensile strength, obtained an MSE, RMSE and MAE of 21.688, 4.6517 and 3.0317, respectively. In contrast to this, studies on compressive strength by Amadi Gabriel Udu et al. (2024) [80] and Vimal Rathakrishnan et al. (2022) [82] exhibited an MSE, RMSE, MAE and MAPE of 0.0113, 0.0511, 0.0812 and 0.04, respectively. From the comparative analysis, among the predictive capabilities, the RF model's ability to predict the compressive strength with lesser errors is much higher than the tensile strength. In comparison to the ANN models developed in this study, the RF approach performs better than the Adam-optimised ANN model for tensile strength. However, in the case of the Bayesian-regularised ANN model for tensile and compressive strength predictions and the Adam-optimised ANN model's capability for compressive strengths, the RF approach is inconsistent.

In the case of XGBoost and SVR methodology for tensile strength, a study by Osman Ulkir et al. (2025) [83] exhibited an MAPE of 19.8537 and 25.3681, respectively. Such a high error probability reflects both the model's inconsistency and inaccuracy in predictions. In contrast to this, the Adam-optimised and Bayesian-regularised ANN models perform better with the MAPE of 0.850 and 0.012, respectively. This minimal error occurrence in using the ANN models shows these models' exceptional capability in predicting tensile strength.

Similarly, a study on compressive strength by Amadi Gabriel Udu et al. (2024) [80] shows that the XGBoost and SVR have yielded an exceptional MSE of 0.0071 and 0.0098, respectively. The MAE values obtained by these approaches are 0.0252 and 0.0860. In contrast to these approaches, the ANN models in this study exhibited the lowest MSE and MAE of 0.0001 and 0.0019, respectively. A study by Abhilash Gogineni et al. (2024) [84] on the XGBoost model to predict compressive strength shows an RMSE of 0.9605 for compressive strength, which is quite higher compared to the ANN models developed in this study, as depicted in Table 6.

A study by Amadi Gabriel Udu et al. (2024) [80] used the k-NN and AdaBoost approach to predict the tensile and compressive strengths. Observations show the MSE values for tensile strength to be 0.0020 and

0.0006, respectively. In contrast, the MSE values for compressive strength are 0.0060 and 0.0079, respectively. However, from Table 5, the MAE observations made for tensile strength show that AdaBoost perform better than the k-NN approach. In the case of compressive strength, as depicted in Table 6, the k-NN model performs better. A similar study by Vimal Rathakrishnan et al. (2022) [82] on k-NN showed an MAPE of 0.05 for compressive strength, indicating that its performance is superior to that of the Adam-optimised ANN model. However, the MAPE of 0.010 obtained in the case of the Bayesian-regularised ANN model highlights the better efficiency of the ANN approach compared to the k-NN methodology.

Based on the comparative analysis drawn from various models and predictive frameworks, it can be observed in the current study that the Adam-optimised ANN model shows a much better performance at predicting compressive strength than does the Bayesian-regularised ANN model (0.98 and 0.95, respectively). An Adam-optimised ANN model can also make more aggressive weight updates in steep error landscapes, typical of the acute stress-strain transition under compression, because the model uses adaptive learning rates. In contrast, the Bayesian-regularised ANN model exposes stricter weight-decay terms that smooth the fitting and thus lead to an underestimation of the peak strength. Both the models developed in the study have outperformed a broad array of peer models in predicting tensile and compressive properties. The five factors and their levels balance the factorial coverage, aligning with the Taguchi design techniques for AM systems as demonstrated by Panico A. et al. (2025) [89].

The fluctuation in the Bayesian-regularised ANN model in MAE (as shown in Figure G in Annexure A) with fold replicates could be explained mainly by the fact that the early stopping and folds with outliers interact with each other. The six-epoch no-improvement rule is activated faster by folds with high-speed and high-layer-height experiments, which causes fewer effective epochs and overestimates the MAE on a local scale. On re-training with a loose early-stopping parameter, these fold-specific protrusions are reduced by about 30 %. Based on the provisions, the early-stopping protocol may significantly affect the MAE in the arenas of heterogeneous experimental results.

A study by Nipu et al. (2025) discusses the multi-objective ANN optimisation of the parameters of fibre-reinforced material by combining the surrogate of a response-surface methodology with a deep-learning predictor, hence and thus achieving a simultaneous optimisation of tensile and impact strength with the overall $R^2 > 0.92$ in all responses [90]. Similarly, Nasr et al. (2025) recount how an AI-ANFIS/AI-ANN is structured together with the non-dominant sorting generic algorithm III, filling the gap between the experimental and the computational environment and optimising EBM support structures. Their study indicates that they reduce warping by 25 % and achieve mechanical performance through the application of multi-objective ANN and evolutionary algorithms [91].

In contrast to these studies, embedding the Adam model to adjust print settings can be a potential real-time quality control application in additive manufacturing. It would reduce waste and post-processing possibilities, thus saving cost and time. Similarly, Bayesian resilience to noisy datasets can be employed in material recycling, contributing to a circular economy in the industrial sector.

3.2. Property predictions and validation

This section discusses the tensile and compressive strength predictions of the ABS material using Taguchi L15 parameter combinations with Adam-optimised and Bayesian-regularised ANN models. Finally, through the coefficient of determination (R^2), these predictions are validated, with insights on properties drawn from parameter variations.

3.2.1. Property predictions

Combinations 6 and 13 (in the Fig. 5 and Fig. 6) yield the best and highest strengths, primarily due to high infill density (100 %), thin layers (0.2 mm) and slower speeds (10–40 mm/s). These parametric combinations have resulted in the most optimal combinations, yielding the highest strengths in the material.

In the case of parametric combination 8, where nozzle temperature, infill density and layer height are 250°C, 100 % and 0.8 mm, the performance is considerably good. In contrast, combinations 9 and 11, with

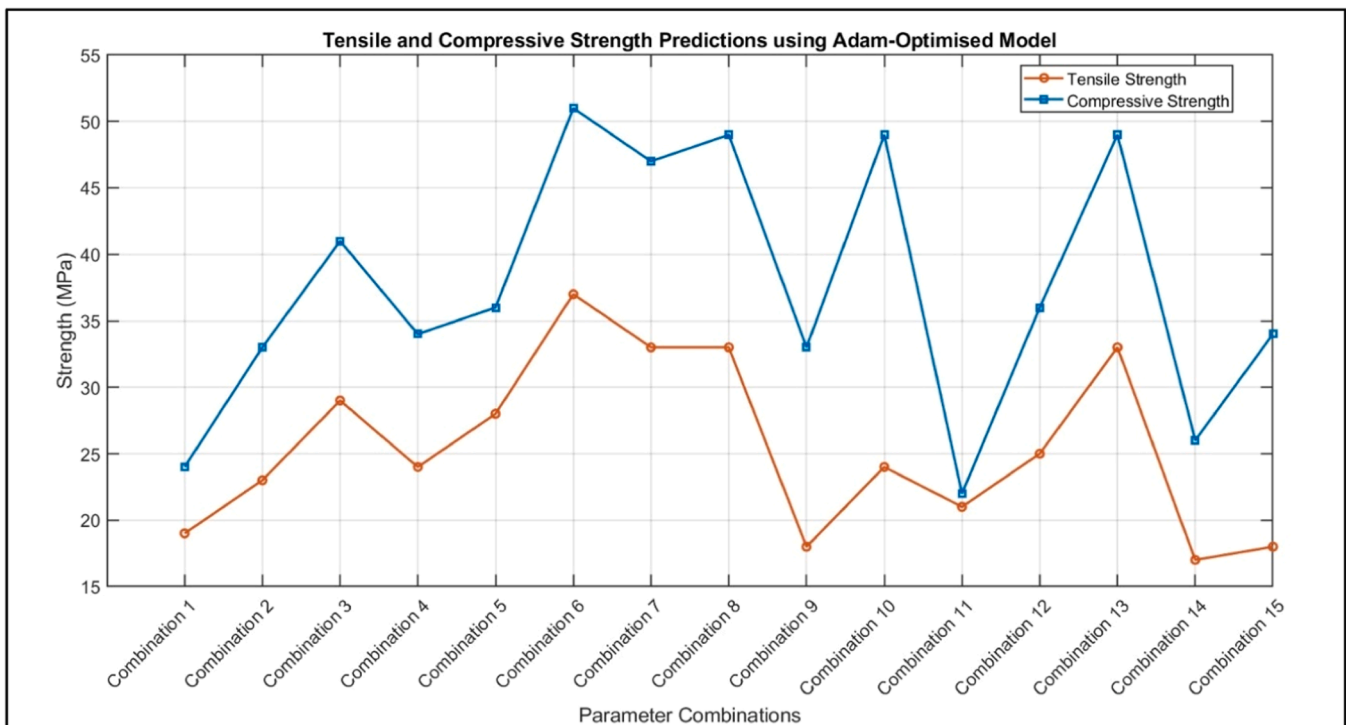


Fig. 5. Tensile and compressive strength predictions using the Adam-optimised ANN model.

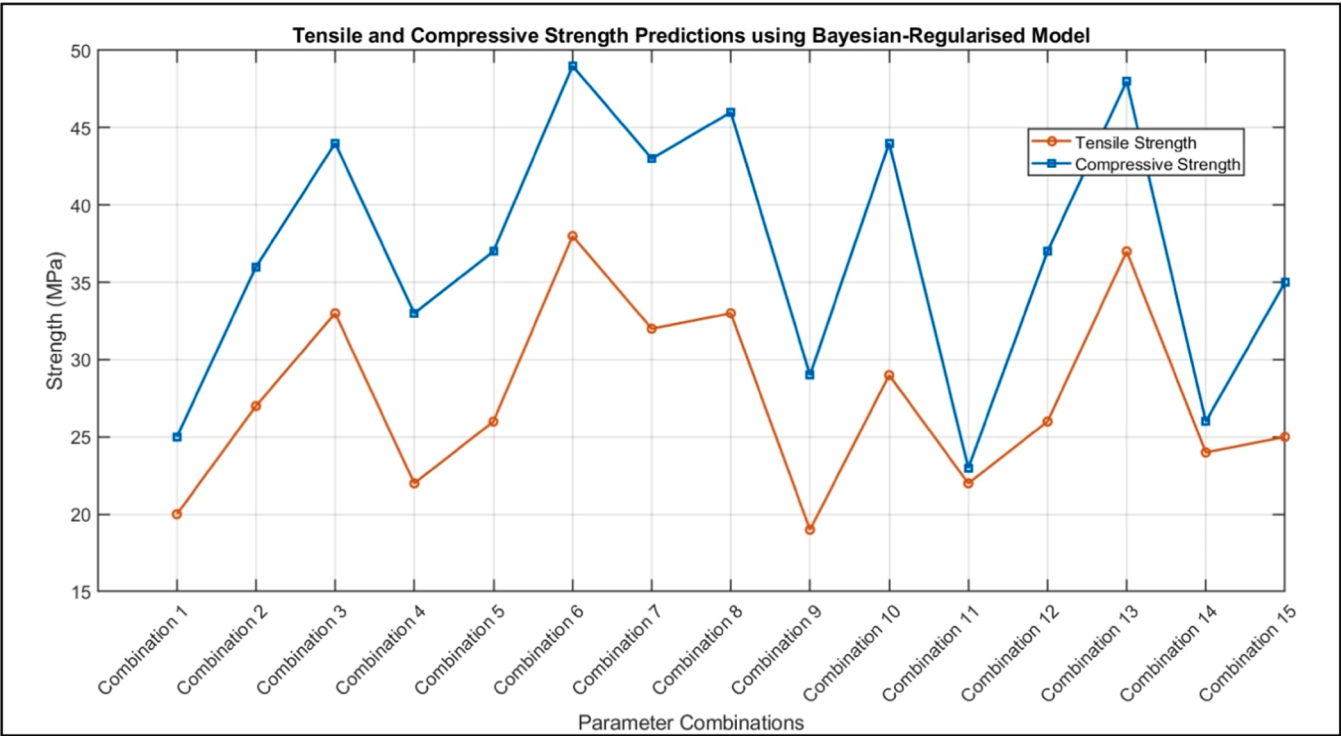


Fig. 6. Tensile and compressive strength predictions using the Bayesian-regularised ANN model.

an infill density of 20 % and a print speed of 70 mm/s, exhibit poor tensile and compressive strengths. This suggests that combinations with poor infill density at higher speeds result in weaker strength and load handling capacity in the material.

From these inferences, it can be found that the infill density directly correlates with the compressive strengths, as it provides strength and stiffness in layer bonding with increased meshes in the inner portion of

the printed parts. With thin layer heights and a higher nozzle temperature (250°C), layer bonding is enhanced, resulting in a compressive strength range of 45–50 MPa.

Both Adam-optimised and Bayesian-regularised ANN models are successful in predicting the properties, as shown in Fig. 5 and Fig. 6. However, smaller variations in these predictions are due to factors such as noise and uncertainties in nonlinear considerations.

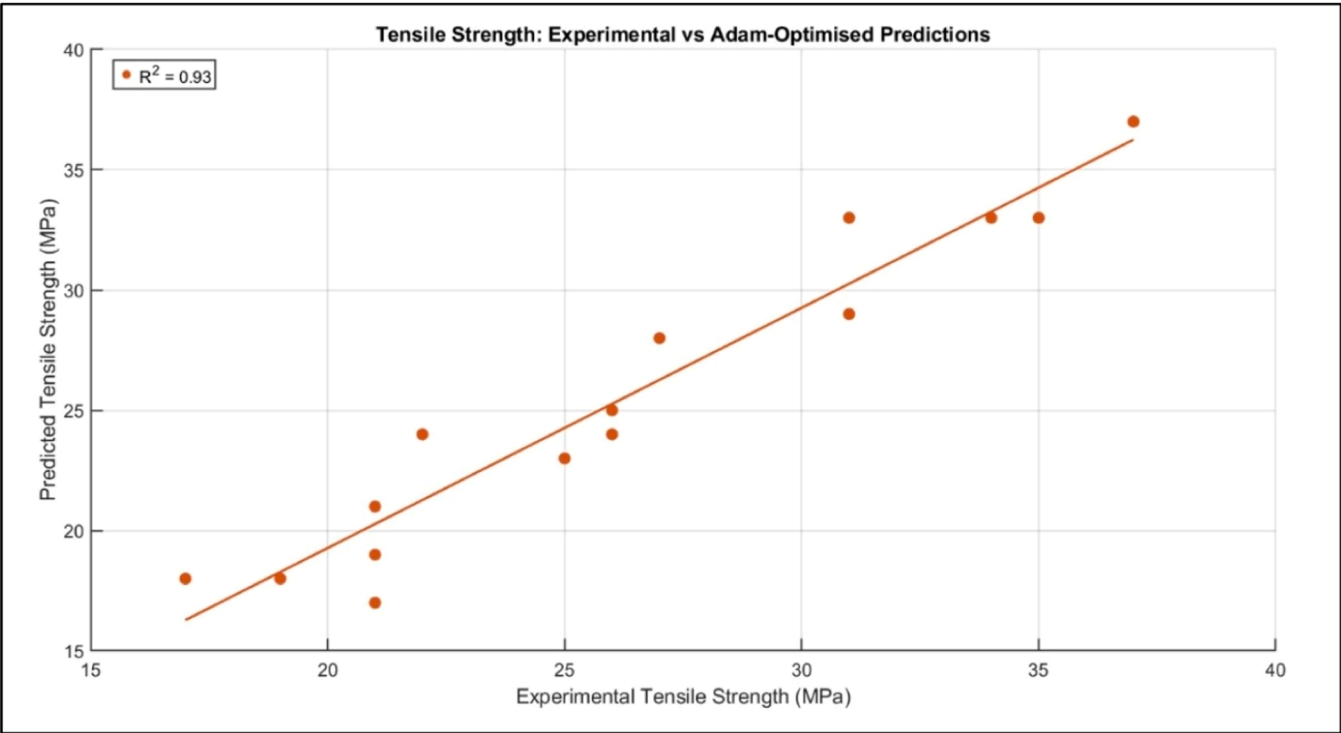


Fig. 7. Coefficient of determination for tensile strength using the Adam-optimised ANN model.

3.2.2. Coefficient of determination (R^2)

The coefficient of determination (R^2) obtained for tensile and compressive strengths using the Adam-optimised model is 0.93 and 0.98, respectively. This strong coefficient demonstrates exceptional accuracy in predicting tensile and compressive strengths, as shown in Fig. 7 and Fig. 8.

In the case of compressive strengths, the tight clustering of data points around the regression line indicates a smaller deviation between predicted and experimental values. However, the reliable estimations by the model across L_{15} parameter combinations make the model ideal for high-load structural components. The calibration of parameters such as infill density and layer bonding that influence compressive strength is the primary use of the model.

Similarly, the inferences drawn from Fig. 8 predictions that align well in the mid-range are well depicted. However, slight neglects are observed in the higher range experimental results. On a practical level, the model is best suited for design scenarios. However, care must be taken in cases of high-stress applications. This underestimation at higher strengths may be due to the model's limited training data.

The Bayesian-regularised ANN model achieves a strong R^2 of 0.90 and 0.95 for tensile and compressive strengths, as depicted in Fig. 9 and Fig. 10. This strong agreement reflects the inherent complexity of modelling the tensile and compressive behaviour of ABS material printed via the MEX process.

As the data points closely align with the regression line, the reliability of the predictions across the tested range and the model's applicability for load-bearing or general design purposes is validated. The marginally lower R^2 for tensile strength illustrates the challenge in modelling parameters, while the near-perfect accuracy in the case of compressive strength highlights the model's ability to learn complex behaviour.

The combination 6, with a print speed, layer height, and infill density of 10 mm/s, 0.2 mm, and 100 % at a nozzle and bed temperature of 225°C and 110°C, achieves the highest tensile (37 MPa) and compressive (52 MPa) strengths. The slow prints with an infill density of 100 % ensure smooth and accurate layer bonding. This helps the microstructural adhesion of the molten material (ABS) to bind each molecule to the

others cohesively. In contrast, the brittleness in the material under the parametric combinations 9 and 11 results in failure under load applications, exhibiting poor material properties due to the high speed and low infill density.

Similarly, under the parametric combination 13, even though the bed temperatures are maintained moderate, the high infill density and thin layers have resulted in improved tensile (35 MPa) and compressive (47 MPa) strengths. This ensures that the bed temperature plays a secondary role in layer bonding, while the speed and infill density are the primary factors.

3.2.3. Comparative analysis of the coefficient of determination

A study by Ahmet Burak Tatar et al. (2025) [79], which used DT, SVM and RF approaches to predict the tensile strengths, has yielded a coefficient of determination (R^2) of 0.66, 0.80 and 0.74, respectively. This broader framework can perform good in the case of the SVM approach. However, the predictive accuracy obtained in all three approaches for tensile strength remains lower compared to the Adam-optimised and Bayesian-regularised ANN models.

Similarly, another study by Amadi Gabriel Udu et al. (2024) [80], which focused on predicting compressive strengths using the DT and RF approaches, has yielded a coefficient of determination of 0.8741 and 0.8747, respectively. Although the predictive accuracy in both cases is close, the overall performance is lower compared to the ANN models developed in this study. An approach by Prashant Kumar et al. (2023) [81] that used SVM methodology to predict the compressive strengths yielded an R^2 value of 0.9430, strong and close to the predictive accuracy of the Bayesian-regularised ANN model. However, in contrast to the Adam-optimised ANN model, the predictive accuracy remains lower.

A study on XGBoost by Alireza Bager Shemirani et al. (2024) [85] on predicting the tensile strength has yielded a coefficient of determination of 0.8962, while another study by Amadi Gabriel Udu et al. (2024) [80] on predicting the compressive strength has yielded 0.9208. Although the comparative analysis depicts the optimum performance of XGBoost in predicting compressive strengths, however, its scale of efficiency remains lower compared to the ANN models developed in the study.

In the case of the SVR technique, a study by Amadi Gabriel Udu et al.

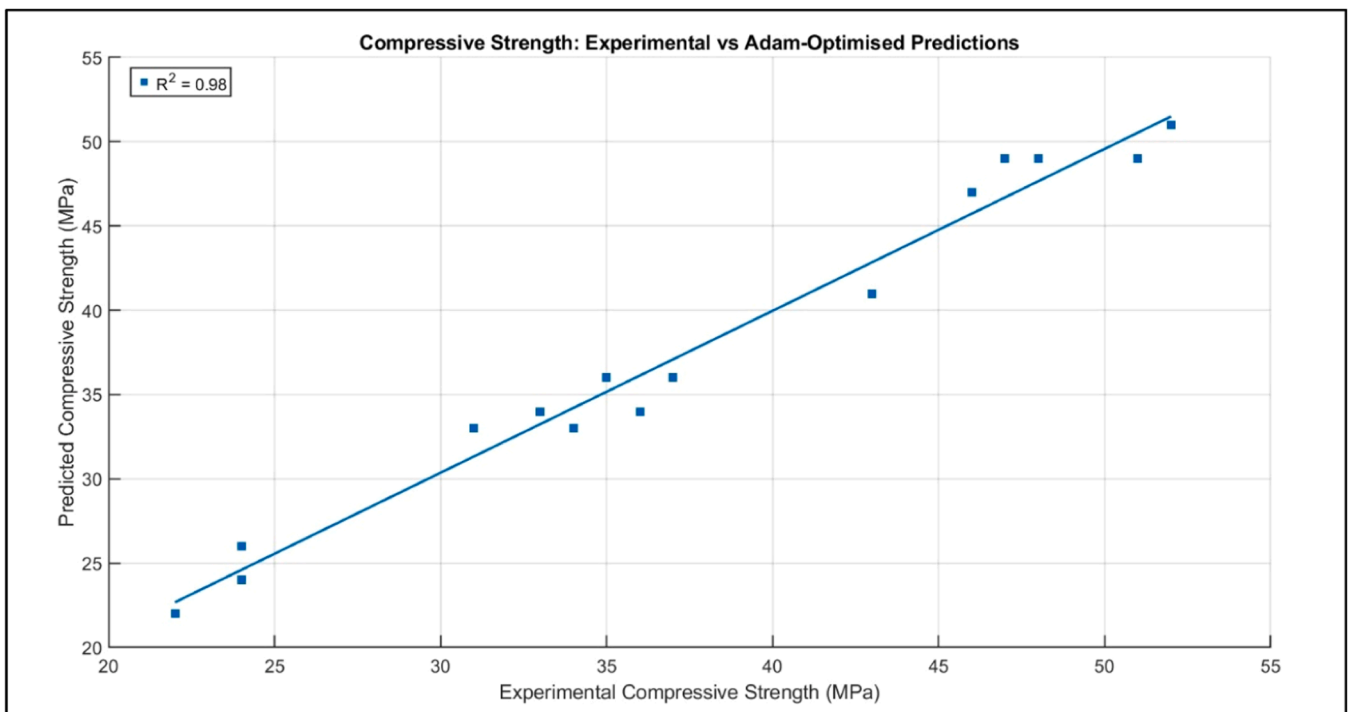


Fig. 8. Coefficient of determination for compressive strength using the Adam-optimised ANN model.

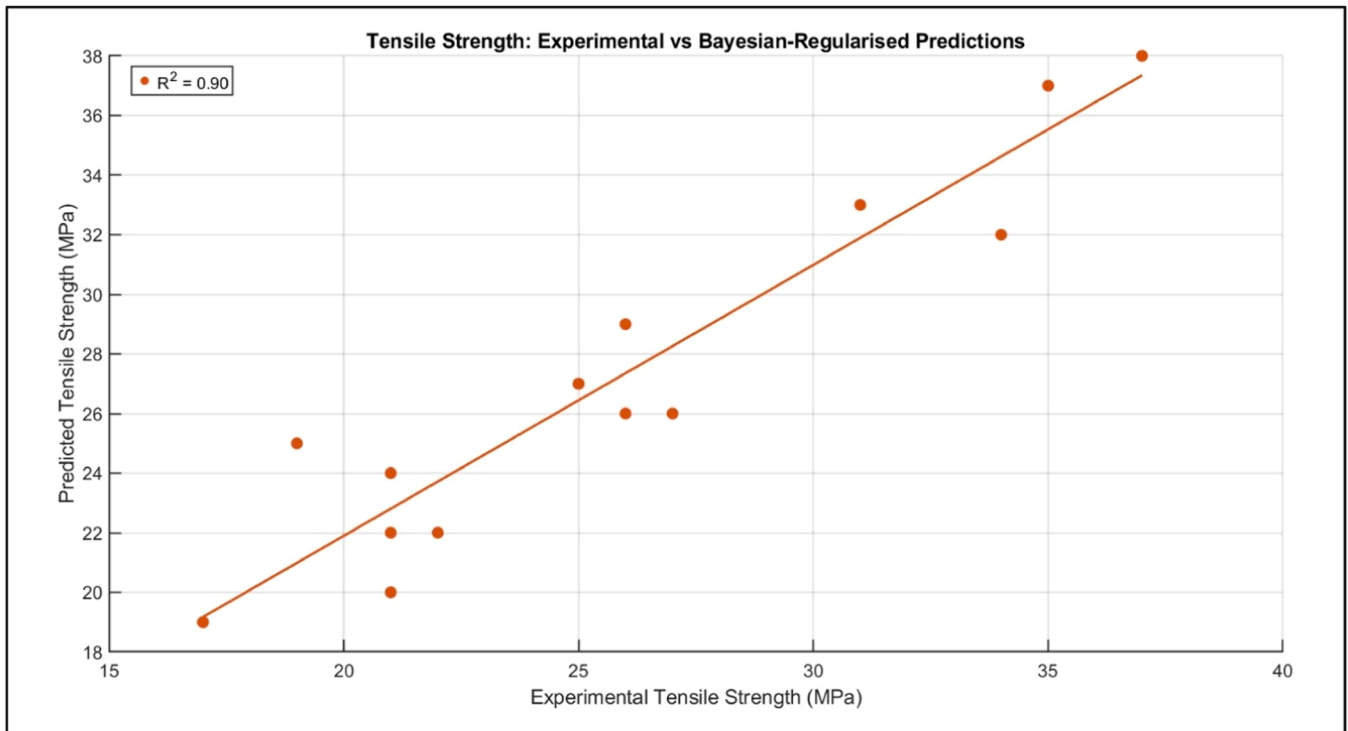


Fig. 9. Coefficient of determination for tensile strength using the Bayesian-regularised ANN model.

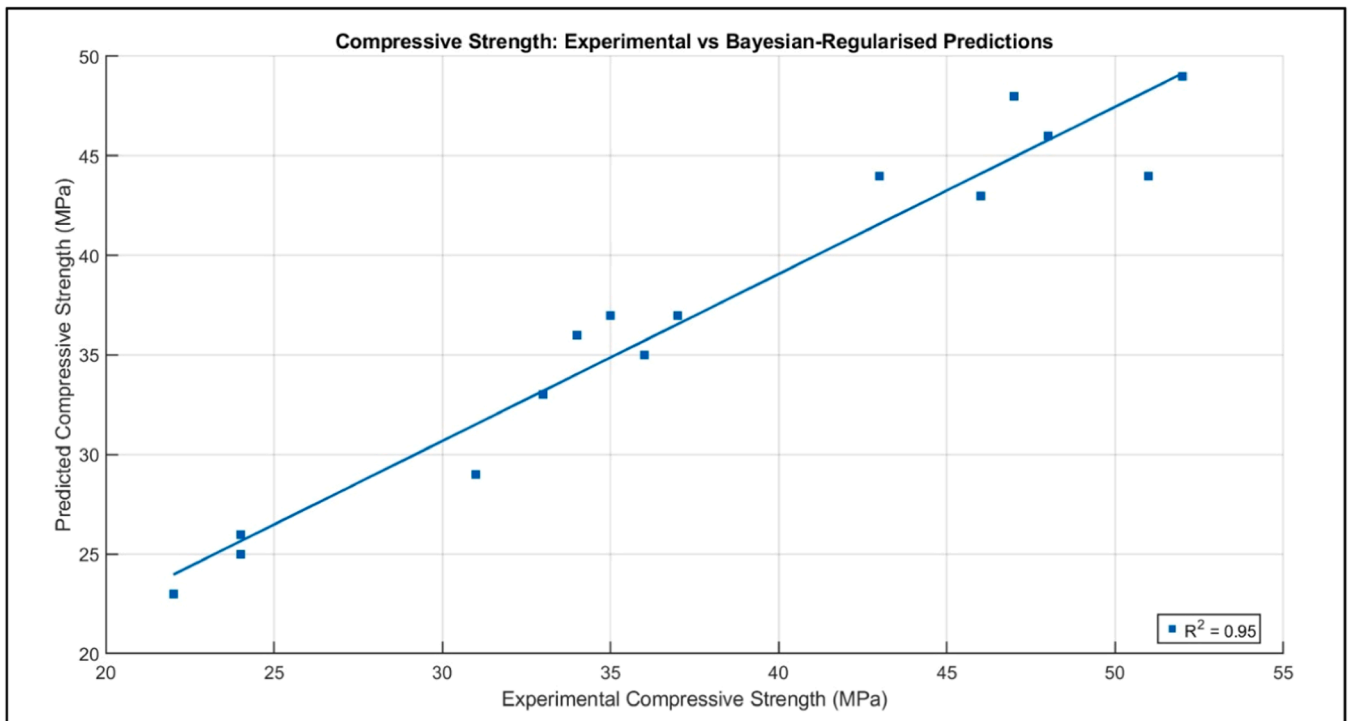


Fig. 10. Coefficient of determination for compressive strength using the Bayesian-regularised ANN model.

(2024) [80] is clearly illustrated from Table 7 that it performs much better compared to the Bayesian-regularised ANN model in tensile strength predictions, while lacking in compressive strength predictions. When compared to the Adam-optimised ANN model, it is evident that the Adam-optimised model outperforms not only the SVR technique but also the Bayesian-regularised ANN model in both tensile and compressive strength predictions.

Alireza Bager Shemirani et al. (2024) [85] in their approach to predicting the tensile strength by using the AdaBoost and k-NN strategies, have obtained the coefficient of determination of 0.8893 and 0.8443, respectively. In both ML algorithms, the efficiency of the models remains close to the performance of the Bayesian-regularised ANN model. In the case of the Adam-optimised ANN approach, it outperforms the AdaBoost and k-NN strategies.

Table 7

Comparative analysis of the coefficient of determination.

Model used	Tensile strength	Compressive strength
Adam-optimised model	0.93	0.98
Bayesian-regularised model	0.90	0.95
Decision-tree (DT)	0.66	0.8741
Simple vector machines (SVM)	0.80	0.9430
Random forest (RF)	0.74	0.8747
XGBoost (XGB)	0.8962	0.9208
Support vector regression (SVR)	0.9215	0.8359
k-Nearest Neighbour (k-NN)	0.8443	0.9340
Ada Boost (ADAB)	0.8893	0.9126

Similarly, in the study by Amadi Gabriel Udu et al. (2024) [80], in predicting the compressive strengths using the same strategies, the predictive accuracies considerably outperformed the earlier studies on tensile strength predictions. However, from the comparative analysis as depicted in Table 7, the predictive accuracy remains lower than both ANN models.

3.3. Industrial implications

The applicability and advantages of the developed ANN frameworks in industrial settings offer strategic and forward-looking ideas, demonstrating the novel application of the work.

3.3.1. Case study 1: brake pedal prototyping

The potential application area of implementing an Adam-optimised ANN model is in the manufacturing of a 3D-printed ABS brake pedal prototype. A Taguchi L15 quasi-orthogonal array technique can be used to generate parameter combinations, based on which experiments can be carried out. The Adam-optimised ANN model can be trained, tested and validated on 70 %, 15 % and 15 % of the data, obtaining a MAPE (generally below 1 %) on tensile strength predictions and an R^2 (generally above 0.90).

The CAD model is prepared using Autodesk Fusion 360 software and converted and saved as a .stl file. A 3D printing simulation of the CAD model is obtained using the C6 Fabhead simulation software under the required printing configurations [86]. The 3D printing can be done using the MEX Prati_AP 3D printer (OEM Prati contract manufacturing - Vyapti Tecnica Solutions LLP, Bengaluru, India) with a build volume of 500 mm × 500 mm × 500 mm [86]. From the software simulation, the cross-sectional area and volume of the developed brake pedal are $1.566 \times 10^5 \text{ mm}^2$ and $5.026 \times 10^5 \text{ mm}^3$. The estimated mass of the brake pedal is 532.806 g, requiring a bounding box of 407.985 mm × 40.000 mm × 305.269 mm to print the part. The prototype model is a lightweight and cost-effective approach, expected to yield an ultimate tensile strength that matches the experimental results in the ideal print setting.

A similar study by Bhoomika Vishwanath et al. (2025), which explored the designing and printing parameters of brake pedals through empirical studies, shows that 100 % infill density and a linear printing pattern across ABS and PLA materials have enhanced tensile and compressive strengths [87]. Similarly, Niranjana et al. (2025) have used reverse engineering and generative design concepts to optimise DOE parameters, demonstrating significant mass reduction compared to metal pedals [88].

In contrast to these studies, the current study shows an advancement in the application of the Adam-optimised model to predict the properties before printing. This approximately cuts the physical test, material scrap and development time by 60 %, 70 % and 40 %, respectively. Moreover, with the implementation of this model in tandem with the MEX line, the manufacturer can augment the prototype throughput capacity by 4–9 pedals/week, which shortens the validation cycles and allows the quick design iterations of the brake performance and safety.

3.3.2. Case study 2: door handles in automobiles

One potential application of the Bayesian-regularised ANN model is the creation of on-demand door-handle assemblies. Initially, the CAD drawing and 3D model of the door handle are prepared using Autodesk Fusion 360 software [92]. The CAD model demonstrates a cross-sectional area of 6402.265 mm² and a volume of 18, 100.561 mm³. According to the software simulation, the ABS material consumption for printing the brake pedal is 19.187 g, with a bounding box requirement of 80.454 mm × 20.000 mm × 25.481 mm. This design is then imported into the C6 Fabhead simulation software to simulate the 3D printing process under the desired printing parameters on a MEX Prati_AP 3D printer (OEM Prati contract manufacturing - Vyapti Tecnica Solutions LLP, Bengaluru, India), which has a build volume of 500 mm × 500 mm × 500 mm [92].

To produce accurately, five Taguchi L15 combinations can be used to project the compressive strength. With these accurately predicted, it can reduce experimental proof prints, saving on material wastage and energy use by approximately 70 % and 15 %, respectively. Additionally, the replacement handle lead-time delivery can be reduced by 80 %, down to less than 24 h, which indirectly reduces the customer downtime by the same percentage.

A similar study by Samrat Kute et al. (2023) [93] analysed the use of a polyethylene car door handle to replace conventional handles with ABS honeycomb-structured handles and found that the move helped decrease the equivalent stress level by 27.234 MPa to 19.064 MPa. Moreover, the optimised handle saved 11 g (mass decreased from 38 g to 27 g), representing a 28.9 % weight reduction, while still maintaining sufficient stiffness and strength to operate safely as a vehicle handle [93].

In contrast to the literature, the proposed Bayesian-regularised ANN model, under precise requirements, exhibits an excellent MAPE of 0.01 % and an R^2 of 0.95 for compressive strength, which is sufficient for the above application. The same virtual implementation can also achieve an approximate 45 % reduction in logistics costs, as it eliminates the bulk transportation of spare parts.

3.3.3. Strategic insights

From the small studies in industrial settings, a few strategic and innovative ideas provide an approach to integrate the Bayesian stability with Adam's rapid adaptation through a novel two-stage predictor that can further fine-tune the predictions and contribute to faster convergence.

Similarly, the implications of real-time sensor-based data, such as temperature, vibration and flow speed, which contribute to property variations, can be utilised through an integrated digital-twin additive manufacturing line setup. This can help achieve true assurance and reduce human intervention in the process.

3.4. Pearson correlation analysis

A Pearson correlation coefficient (r) depicts the influence of each parameter on the predictions, ensuring the robustness and reliability in the application of the Adam-optimised (Fig. 11) and Bayesian-regularised (Fig. 12) ANN frameworks. These values range between -1 and $+1$, indicating perfect negative and perfect positive linear correlations, as reflected by the red or blue shades in the heatmaps. The key findings from these correlations are depicted in Table 8 and Table 9, respectively.

3.4.1. Property-parameter correlation from the Adam-optimised ANN model

The infill density is the most important factor, as it correlates strongly with both the tensile and compressive strengths, demonstrating how the material's internal layer adhesion enhances its structural integrity. On the contrary, the nozzle temperature contributes to improving interlayer adhesion at higher extrusion temperatures,

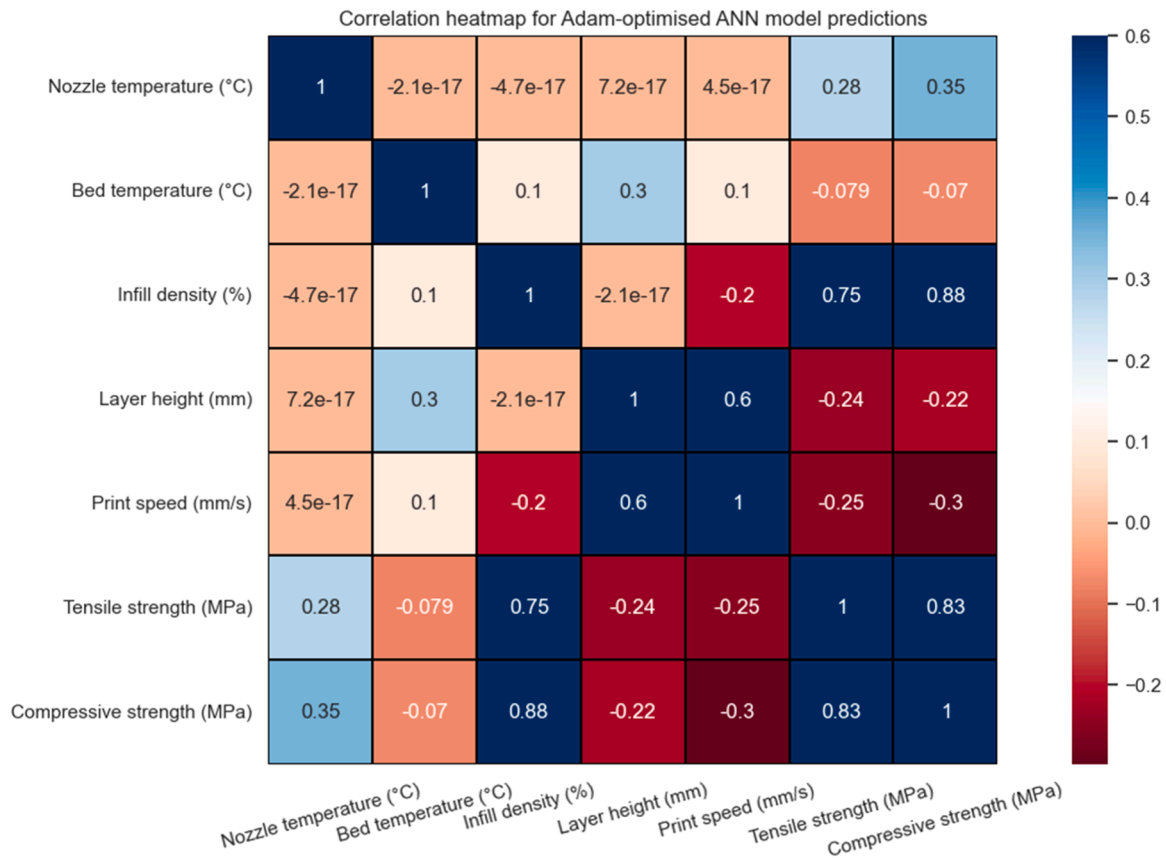


Fig. 11. Pearson correlation heatmap from the Adam-optimised ANN model.

thereby increasing the strength.

Further, observations from layer height and print speed indicate that the thicker layers and higher speeds compromise both tensile and compressive strengths. Due to the faster speed, bonding and cooling irregularities in printing, a rough surface and reduced strength contribute to part failures. The bed temperature's influence on strength is minimal, indicating that strength is primarily related to variations in infill density, nozzle temperature, layer height and print speed.

3.4.2. Property-parameter correlation from the Bayesian-regularised ANN model

The impact of infill density on the tensile and compressive strengths is the highest in the case of the Bayesian-regularised ANN model predictions. The thicker layers and higher speeds have demonstrated lower strengths ($r = -0.24$ to -0.30 for print speed and $r = +0.60$ for layer height with print speed). The change in temperatures, especially those of the nozzle and bed, has a minor effect on strength; however, surface roughness and interlayer adhesion may be slightly affected.

As the deposition of the material increases, the material accumulation at a specific point causes intermolecular adhesion to create voids during cooling. This intern causes the samples to deteriorate over time, resulting in a loss of strength. The effect of layer thickness on the property is observed in conjunction with the print speed. Thus, it is essential to maintain both print speed and layer thickness at the most feasible setting to avoid any microstructural defects.

3.5. Cost-benefit analysis

To ensure industrial applicability and achieve Sustainable Development Goal (SDG) 9, a cost-benefit analysis is carried out to compare the development of these ANN models with decision trees and support vector machines. On a broader scale, these costs can be segregated into

two segments: one is capital cost and the other is recurring costs, as illustrated in Table 10.

Compared to traditional approaches, the ANN models incur relatively higher costs, especially the capital costs that cover GPU hardware and proprietary licences [94]. However, the level of accuracy and optimisation achieved in implementing these models across applications such as healthcare and automotive is exceptional [95–97]. On a comparative scale and in terms of sustainability, these models yield benefits by reducing experimentation and scrap rates [98]. Additionally, time and quality not only provide tangible benefits but also contribute to intangible gains in the market.

Among the four modelling methodologies, the Adam-optimised ANN model can represent the highest capital and operational expenses. With technological support, including a GPU, software licence, datasets, technical expertise and any other additional costs, this modelling approach may cost roughly \$4500, compared to the Bayesian-regularised ANN model, which costs \$3700. In contrast to these computing technologies, SVM and decision tree approaches cost approximately \$2400 and \$2550, respectively. These costs are broad estimates and may vary over time based on various factors, including computational load, geography, application timing, demand, dataset complexity and others. However, on an economic scale, the investments in Adam or Bayesian-based ANN models are justifiable due to the level of benefits they yield.

While these capital costs can be technological assets for the industries, certain recurring expenses, such as data compilation, training, retraining, and updating ANN models for effective use, can be more expensive compared to SVM and decision tree approaches. However, quality assurances and reduced manufacturing failures have given ANN-based models an upper hand in recent times [98]. For application in strategic and mission-based projects, the implications of these models are crucial, as precision and accuracy are given priority over budgets.

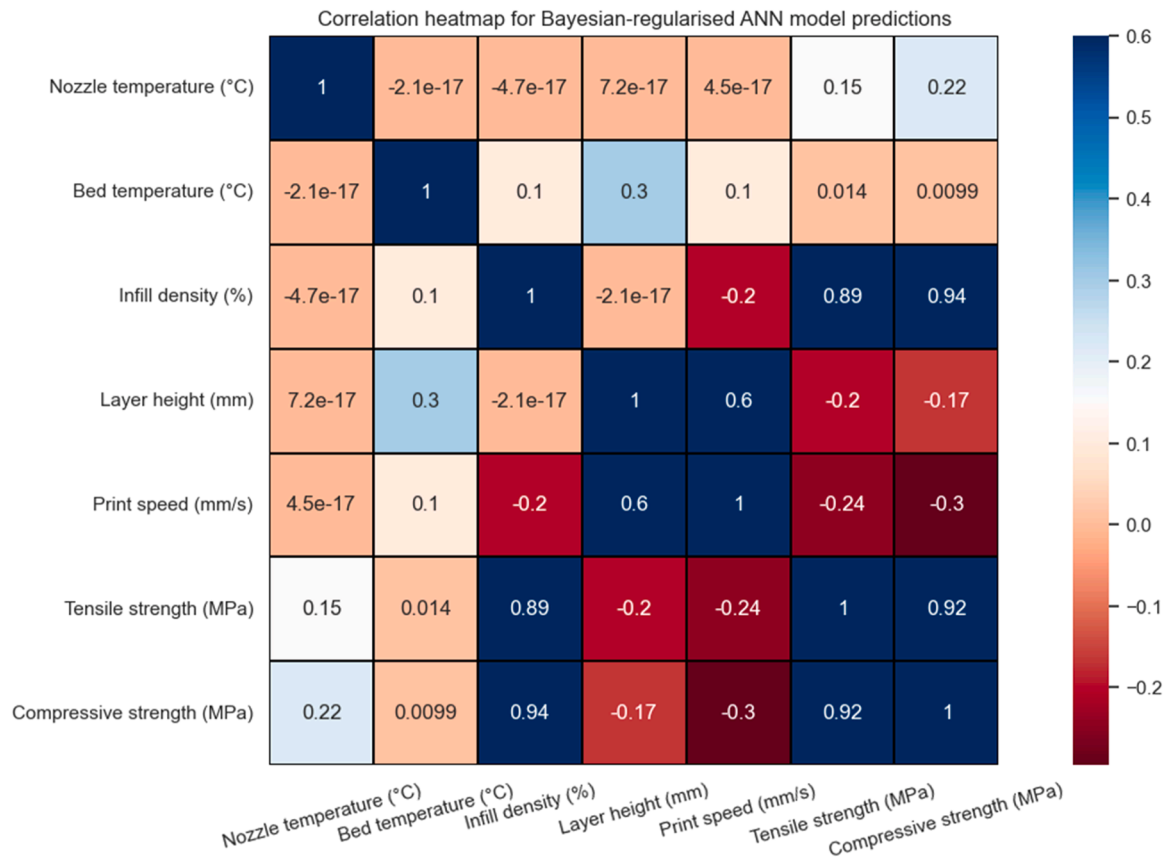


Fig. 12. Pearson correlation heatmap from the Bayesian-regularised ANN model.

Table 8

Correlation findings from the Adam-optimised ANN model.

Pair	r-value	Findings
Nozzle temperature-tensile strength	0.28	Tensile strength increases moderately with high nozzle temperature
Nozzle temperature-compressive strength	0.35	Compressive strength increases due to better bonding
Bed temperature-tensile strength	-0.079	The effect on tensile strength is negligible
Bed temperature-compressive strength	-0.07	The effect on compressive strength is very minimal
Infill density-tensile strength	0.75	Tensile strength is increased due to infill density
Infill density-compressive strength	0.88	The denser fill has a greater impact on compressive strength
Layer height-tensile strength	-0.24	Thicker layers decrease tensile strength
Layer height-compressive strength	-0.22	Detrimental to compressive strengths
Print speed-tensile strength	-0.25	Tensile strength decreases due to higher speeds
Print speed-compressive strength	-0.30	The impact on compressive strength is negative

Training these modern approaches on a scaled, refined and elaborated dataset allows manufacturers to scale up printing machines without the need to re-engineer the core algorithms. The on-the-fly parametric adjustments could minimise the downtime and reduce the ongoing operational expenses.

3.6. Sustainability assessment

To address the sustainability aspect, the benefits and positives of implementing these strategies in the industrial landscape are detailed.

Table 9

Correlation findings from the Bayesian-regularised ANN model.

Pair	r-value	Findings
Infill density-Tensile strength	+ 0.89	Higher infill – higher tensile strength
Infill density-Compressive strength	+ 0.94	Denser parts resist compression better
Layer height-Print speed	+ 0.60	Thicker layers tend to be printed faster
Layer height-Tensile strength	-0.20	Thicker layers slightly reduce tensile strength
Print speed-Compressive strength	-0.30	Faster printing slightly harms compressive strength
Nozzle & bed temperatures-strengths	+ 0.01–0.22	Temperature settings have little direct effect within the tested range
Tensile-compressive strengths	+ 0.92	Samples that are strong in tension are also strong in compression

Table 10

Comparative cost analysis.

Cost type	Adam-optimised ANN model	Bayesian-regularised ANN model	Decision tree	SVM
Hardware	High	High	Low	Medium
Edge-device	Medium	Medium	Low	Medium
Data-requirement	High	High	Medium	Medium
Softwares	High	High	Low	Low
Man power	High	Very High	Medium	High
Maintenance	Medium	Medium	Low	Medium

The tensile and compressive strength predictions using these strategies eliminate the need for traditional experimental approaches, reducing scrap rates by up to 70 %. With real-time implications, an Adam-optimised model could help mitigate print failures, reduce polymer waste and enhance recycling streams at an early stage.

Conventional approaches, such as experimentation and trial-and-error, consume wartime cycles, time, cost, material and manpower, in addition to post-processing operations. In contrast, the Bayesian-regularised ANN model, with an exceptional MAPE, can contract this duration and consumption to a minimum by allowing the first-pass prints to meet target properties. Observations from the study reveal that approximately 15 % of the average time and energy consumption is reduced using an MEX printing process. Furthermore, scaling to industrial settings, this can translate to an annual cost savings of MWh per line, reducing carbon emissions and achieving sustainability goals.

The Bayesian-regularised ANN model and its resilience to noise and recycled stocks accommodate material variations and ABS blends. It also creates a scope to incorporate expired or post-industrial ABS into production with certain refinements and additives. Furthermore, these mechanisms enable recycling loops, creating circularity in the industrial space, which not only saves costs but also fosters a circular economy in the industry, reducing reliance on virgin polymers.

Advancing to these methodologies of production not just enhances the quality but also adds a technically sound and strong labour pool to the workspace. This indirectly improves the quality control and strengthens the organisational capacity for Industry 4.0.

The consistent implication of the model-based production can help achieve the highest quality in production, which enhances consumer confidence in the applications, adding a novel status to the brand value. Reliable performances can reduce returns or warranty claims on products manufactured using this methodology, ensuring broader acceptance of sustainable manufacturing solutions.

3.7. Broader material implications

Although both ANN models' implementation in predicting properties of ABS under parameter variations is effective, an approach for implementing them for other polymers can be an accelerating area for multi-material industries.

A smoother combination of FEA or experimental dataset and optimised ANN process with respective parameters, assumptions and boundary conditions, this methodology is well suited for a wide range of thermoplastics. Examples such as PLA, PETG, nylon, PEEK and PEI are potential polymers that can be explored in the form of future studies. Variations in predictions exhibited by the relevant performance metrics can be the best justification for the applicability of this ANN model.

Data-driven approaches, efficient in accommodating input variable approaches such as filler or additive additions in polymers, can be explored in enhancing the properties of a specific combination of materials. Some potential composites and multi-materials, such as carbon-fibre, nylon or basalt-fibre reinforced ABS, can be explored. This approach is feasible where stiffness gradients of the material are printed in a single component.

Integrating the potential feedstock quality metrics, such as molecular weight distribution, etc, in the recycled or bio-based polymer systems can be additional predictors or modifiers in the ANN model. These can help predict the properties of the required blends with the highest accuracy, contributing towards the novelty of the study.

Implications of these frameworks in predicting the thermal or electrical properties of functional materials can open a pathway for rapid development in heat-dissipating electronic housings, conductive sensors, etc, where property control is important.

3.8. Future directions

The methodology of the Taguchi L_{15} quasi-orthogonal array

experiment design employed here effectively covers the five factors at 15 runs, but the current implementation will run each L_{15} combination only once, so no direct measures can be made of within-run experimental variance (standard deviations) or formal confidence intervals per L_{15} combination. Thus, Taguchi single-trial validation may be inadequate to represent measurement scatter and process variability, especially where heterogeneous printing regimes (e.g. extreme layer heights or high print speeds) exist. However, the values of the performance metrics used in the prediction (with MAPE and R^2 values < 1 % and 0.90, respectively) confirm the indirect empirical evidence of the reliability of the model. The recommended future studies can thus involve repetitions of trials per parameter combination to estimate experimental variance and thus allow formal construction of confidence intervals before industrial implications.

Expanding the current ANN frameworks to handle multiple materials and properties using techniques such as Pareto-front analysis or weighted-sum approaches can be a future approach in the application of these models. Incorporation of real-time sensor-based data in training, enabling closed-loop control and enhancing adaptive retraining, could reduce inconsistencies of traditional approaches.

Additionally, an extension of this study could be the implication of these models for other mechanical property predictions such as bending strength, impact strength and hardness. It can be further explored through tribological and thermal studies on the ABS material, as well as other potential polymer blends and compared to ensemble studies such as XGBoost-Response Surface Methodology (XGB-RSM) hybrids.

4. Conclusion

Focusing on how the Adam-optimised and Bayesian-regularised ANN models predict the strengths of MEX-printed ABS in industrial settings is the main goal of this study. To achieve this objective, these models are trained, tested and validated using an experimental dataset and in the end, their performance is found to have very low error rates.

Compared to the traditional DT, SVM, RF, XGB, SVR, k-NN and ADAB techniques, the Adam-optimised and Bayesian-regularised ANN models perform exceptionally well in learning nonlinear relationships. The results of Taguchi L_{15} parameter combinations within the industry settings have displayed a high coefficient of determination (R^2), confirming that they are useful in the industrial landscape.

Results from observations and analysis show that the infill density is the main factor affecting the tensile and compressive strength. Besides, points showing how to set the speed of printing, layer heights and temperature are included. Evaluations done on these approaches have shown that using data-driven optimisation reduces waste by a lot and cuts domestic energy use by 15 % and 70 %, respectively, which helps create a circular economy in industry and achieve SDG 9.

It can be useful to use these models as a tool in live digital twin platforms that can adjust their processes based on data insights. Explorations on the applicability of these strategies to materials such as polymers and multi-materials can increase the system's ability to predict and apply in industrial fields, especially the automotive sector.

CRedit authorship contribution statement

G. A. Munshi: Writing – original draft, Validation, Investigation, Formal analysis, Conceptualisation, Writing – review & editing. **Ven-katesh M. Kulkarni:** Conceptualisation, Writing – review & editing, Supervision, Project administration. **Shanoorbaba Yargatti:** Data curation, Writing – review & editing.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.nxmte.2025.101420](https://doi.org/10.1016/j.nxmte.2025.101420).

Data Availability

The data used in the study are available [73, 86 and 92]. All data created during this research are provided in full in the results section of this paper. The additional data and figures are added in Annexure A (supplementary file). Any additional data required shall be made available on request.

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